

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM (SECMS)

Salesforce-Based Academic Enrollment Management Application

Project Documentation

Salesforce Configuration • Flows • Approval Processes • Reports • Dashboards • Security

1. INTRODUCTION

1.1 Project Overview

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring

2. IDEATION PHASE

2.1 Problem Statement

PS	Customer	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollments efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone
PS-2	Administrator	track enrollments and revenue	reporting is delayed	data is not centralized	uncertain in decision-making

2.2 Empathy Map Canvas

Area	Details
User Persona	Enrollment Officer
Think & Feel	Concerned about double bookings and revenue loss
Hear	Complaints from students and management
See	Multiple spreadsheets and emails
Say & Do	Rechecks data, follows up manually
Pain	Duplicate entries, approval bottlenecks
Gain	Automation, dashboards, faster processing

2.3 Brainstorming & Idea Prioritization

Step 1 - Brainstormed Ideas

- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 - Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 - Prioritization

Salesforce-based SECMS was selected due to high feasibility and high impact.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages covered: Awareness & Registration → Course Selection → Enrollment Request → Fee Payment → Approval → Instructor Assignment → Reporting & Completion

3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability
Simple, intuitive, user-friendly interface with clearly labeled tabs, dashboards, and forms. Examples: quick student registration form, one-click enrollment approval, role-based dashboards.

NFR-2: Security
Confidentiality, integrity, authentication, authorization, and access controls. Examples: only admins modify course details; instructors view assigned students; students cannot access other student records.

NFR-3: Reliability
Consistent and accurate operation during enrollment, approvals, and reporting. Examples: no duplicate registration, accurate status updates, consistent reports.

NFR-4: Performance
Fast response times, especially during peak admission periods. Examples: quick dashboard refresh, fast student search, no lag during bulk uploads.

NFR-5: Scalability
Support growth in students, courses, departments, and users and allow addition of new modules.

NFR-6: Availability
Continuous accessibility during working hours with minimized downtime and reliable cloud access.

3.3 Data Flow Diagram

User inputs student and enrollment data → System validates and stores records → Approval process routes data → Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem-Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

4.2 Proposed Solution

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Agile methodology • Epics → Stories → Story Points • Sprint-based execution • Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system.	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently.	3	High	Member2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately.	5	High	Member4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High	Member4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium	Member1
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry.	5	High	Member2
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster.	3	High	Member4
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High	Member1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High	Member2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High	Member3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can quickly analyze system performance.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Completed	Actual Release
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. PROJECT DEVELOPMENT PHASE

Project Flow:

- Milestone 1: Salesforce Developer Account Creation
- Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)
- Milestone 3: Tabs Creation
- Milestone 4: Lightning App Creation
- Milestone 5: Fields Creation
- Milestone 6: Create Relationships
- Milestone 7: Page Layout Customization
- Milestone 8: Validation Rules

- Milestone 9: Flows (Automation)
- Milestone 10: Approval Process
- Milestone 11: Reports
- Milestone 12: Dashboards
- Milestone 13: Security Setup
- Milestone 14: Conclusion

What You'll Learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships
- Creating page layouts & record types
- Writing validation rules
- Building automation with flows
- Creating approval workflows
- Developing reports & dashboards
- Setting up security with profiles, permission sets, and sharing
- End-to-end application testing

Milestone 1 - Salesforce

Introduction

Salesforce is a customer success platform designed to help organizations sell, service, market, analyze, and connect with customers. It provides standard products and features to manage relationships, collaborate with employees and partners, and securely store data in the cloud.

Reference video: Salesforce introduction (YouTube link supplied in source document).

Activity 1: Creating Developer Account

Go to the Salesforce Developer signup page and enter: First name & Last name, Email, Role: Developer, Company: College Name, Country: India, Postal Code: PIN code, and a username combining name and company. Click "Sign Me Up" after filling the form.

Developer signup page: <https://developer.salesforce.com/signup>

Activity 2: Account Activation

- Open the inbox of the email used while signing up and click the verification email.
- Click Verify Account.
- Create a password, answer the security question, and click Change Password.
- You will then be redirected to the Salesforce Setup page.

Milestone 2 - Objects

Activity 1: Creating Student Object

Purpose: Store and manage student personal details, contact information, and enrollment status.

- Go to Setup → Object Manager.
- Click Create → Custom Object.

- Label: Student; Plural Label: Students; Object Name: Student.
- Record Name: Student Name; Data Type: Text.
- Enable Allow Reports, Allow Activities (optional/recommended), Track Field History (optional), and Allow Search.
- Click Save.

Create Courses, Instructors, and Enrollments objects using the same general procedure.

Milestone 3 - Fields

Student Object Field	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Course Object Field	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Instructor Object Field	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Enrollment Object Field	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 - Create Relationships

Activity 1: Create Lookup from Enrollment to Student

Purpose: Link each Enrollment to a Student.

- Setup → Object Manager → Enrollment__c → Fields & Relationships → New.
- Select Lookup Relationship → Next.
- Related To: Student__c → Next.
- Field Label: Student (API Name: Student__c).
- Complete Field-Level Security and Page Layout options → Save.

Activity 2: Create Lookup from Enrollment to Course

Purpose: Link each Enrollment to a Course.

- Enrollment__c → Fields & Relationships → New → Lookup Relationship.
- Related To: Course__c.
- Field Label: Course (API Name: Course__c).
- Complete visibility/layout settings → Save.

Activity 3: Create Lookup from Enrollment to Instructor

Purpose: Store the Instructor assigned to an Enrollment.

- Enrollment__c → Fields & Relationships → New → Lookup Relationship.
- Related To: Instructor__c.
- Field Label: Instructor (API Name: Instructor__c).
- Complete Field-Level Security and Page Layout options → Save.

Milestone 5 - Validation Rules

Activity 1: Student Email Domain Validation

Purpose: Ensure students use only allowed institutional email addresses.

- Go to Setup → Object Manager → Student__c → Validation Rules → New.
- Rule Name: Student_Email_Domain_Validation.

Error Condition Formula:

```
NOT(CONTAINS(Email__c, "@student.edu"))
```

Error Message: Email must be a valid student institutional email. Error Location: Field → Email__c.

Activity 2: Fees Must Be Paid Before Approval

Purpose: Prevent Enrollment from being approved until fees are paid.

Error Condition Formula:

```
AND(<br/> ISPICKVAL(Enrollment_Status__c, "Approved"),<br/> Fees_Paid__c = FALSE<br/>)
```

Error Message: Fees must be paid before approving an enrollment. Error Location: Top of Page (or field as needed).

Milestone 6 - Approval Process

Activity 1: Create Approval Process for Enrollment Requests

Purpose: Route Enrollment requests to the Training Manager for approval or rejection.

- Go to Setup → Approval Processes → select Enrollment__c.

- Click Create New Approval Process → Use Standard Setup.
- Process Name: Approval_Enrollment_Request.
- Entry Criteria: Enrollment_Status__c = Requested.
- Add approval step with the designated Training Manager as approver.
- Final Approval Action: Field Update → Enrollment_Status__c = Approved.
- Final Rejection Action: Field Update → Enrollment_Status__c = Rejected.
- Add an Email Alert for rejection notification.
- Activate the approval process.

Milestone 7 - Tabs

Create Custom Object Tabs for Student, Courses, Instructors, and Enrollment through Setup → Tabs → Custom Object Tabs → New. Select the object, choose a tab style, assign profiles, and save.

Milestone 8 - The Lightning App

Create a Lightning application named “Student Enrollment & Course Management”.

- Setup → App Manager → New Lightning App.
- Set App Name and optional description/branding.
- Navigation Style: Standard Navigation.
- Assign the app to appropriate profiles such as Admin, Enrollment Officer, and Instructor.
- Include Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, and Tasks.
- Review and click Save & Finish.

Milestone 9 - Page Layouts and Record Types

Activity 1: Create Record Types for Enrollment

- Enrollment__c → Record Types → New.
- Create New Enrollment from Master and enable it for required profiles.
- Create Re-Enrollment similarly.
- Assign appropriate default page layouts.

Activity 2: Separate Page Layouts

- Create New Enrollment Layout by cloning the existing Enrollment Layout.
- Keep fields relevant to first-time enrollment.
- Create Re-Enrollment Layout and include previous enrollment reference/reason where applicable.
- Use Page Layout Assignment to map each record type to its layout.

Activity 3: Enrollment Related Lists

Add the Enrollments related list to the Student, Course, and Instructor page layouts so users can see related enrollment records.

Milestone 10 - Automation with Flow

Activity 1: Auto-Populate Enrollment Date

Create a Record-Triggered Flow on Enrollment__c when a record is created. Add Update Records to set Enrollment_Date__c to \$Flow.CurrentDate. Save as SetEnrollmentDate and Activate.

Activity 2: Send Email When Enrollment Approved

Create an email template and alert. Create a Record-Triggered Flow on Enrollment__c update with condition Enrollment_Status__c = Approved. Get the related Student, send the email alert, optionally notify Admin, and update Student status to Active. Save as Enrollment Approved Action and Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Prepare Instructor records with codes such as INSTR_A, INSTR_B, and INSTR_C. Create a flow on Enrollment__c, get the related Course, use a Decision on Course Category, retrieve the matching Instructor, and update Instructor__c. Rules: Technical → Instructor A; Language → Instructor B; Non-Technical → Instructor C.

Activity 4: Auto-Create Task for Follow-Up

Create a Record-Triggered Flow on Enrollment__c when Enrollment_Status__c = Requested. Get the Admin user, calculate TODAY() + 2 for the due date, and create a Task with subject “Follow up on enrollment request”, OwnerId as Admin, WhatId as Enrollment Id, and ActivityDate as the calculated date.

Milestone 11 - Reports

Students by Status

Report Type: Students; group rows by Status__c; add Name, Email, and Phone.

Enrollments by Course

Report Type: Enrollments; group rows by Course__r.Name; show record count per course.

Pending Enrollment Approvals

Enrollments report; filter Enrollment_Status__c = Requested; add Student, Course, Owner, and Created Date.

Revenue Report (Fees Paid)

Enrollments report; filter Fees_Paid__c = TRUE; group by Course__r.Name; summarize the applicable fee field.

Milestone 12 - Dashboards

Create a dashboard named “Student Management Dashboard”. Add: Donut Chart using Students by Status; Bar Chart using Enrollments by Course; Gauge using Revenue Report – Fees Paid; and Table using Pending Enrollment Approvals. Arrange components and click Save → Done.

Milestone 13 - Security Setup

Activity 1: Configure Profiles

Profile	Suggested Access
Admin	Full access
Enrollment Officer	Create/Update Enrollments; Student/Course access as required
Instructor	Read-only access to assigned Enrollments

For Enrollment Officer, configure Enrollment__c with Read, Create, and Edit permissions and Student__c/Course__c with Read (and Edit if required). Repeat with appropriate read-only access for Instructor.

Activity 2: Create Permission Sets

Create permission sets for additional object access and dashboard/report editing where needed, then assign them to relevant users.

Activity 3: Configure Sharing & OWD

Example OWD configuration: Students = Public Read/Write; Courses = Public Read/Write; Enrollments = Private. For Instructor access to assigned Enrollments, an Instructor User lookup can be used and a Record-Triggered Flow can set OwnerId to the Instructor User.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the functioning of implemented features of SECMS. These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; record type behavior; flow execution for automated approvals and seat availability checks; Apex trigger functionality where implemented; and reports and dashboards for enrollment tracking and performance monitoring. Testing screenshots should be submitted as project evidence.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Centralized data	Salesforce dependency
Automated workflows	Requires initial configuration effort
Real-time reporting	

9. CONCLUSION

The Salesforce Student Enrollment & Course Management System successfully streamlines the student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce automation tools such as validation rules, flows, approval processes, and dashboards, the project improves data accuracy, operational efficiency, and decision-making. The project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects

- [GitHub / Demo: Salesforce Developer Org](#)

THANK YOU